Combinatorics 3

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Problem 1

With PIE:

Let A_i be the set of such functions where $f(i) = i$. Then we want to count

$$\left|\bigcap_{i\in[n]} \overline{A_i}\right| = n^n - \left|\bigcup_{i\in[n]} A_i\right|.$$

We have that $|A_i| = n^{n-1}$, since we can fix one element, i, and the remaining can map anywhere. Extending this, we have that $|A_{i_1} \cap \cdots \cap A_{i_k}| = n^{n-k}$. The number of ways to choose the k indices is $\binom{n}{k}$, so using PIE,

$$n^n - \left|\bigcup_{i\in[n]} A_i\right| = n^n - \sum_{k=1}^{n}(-1)^{k-1}\binom{n}{k}n^{n-k} = \sum_{k=0}^{n}(-1)^{k}\binom{n}{k}n^{n-k}.$$

Without PIE:

We assign each element of $[n]$ to any other element of $[n]$ except itself, giving $(n-1)^n$.

Problem 2

We can count the number of ways to choose a hand of 13 cards that only contains 2-10 and subtract the result from $\binom{52}{13}$. Since there are $9 \cdot 4 = 36$ such cards, the result is $\binom{52}{13} - \binom{36}{13}$.

Problem 5

If the excedance set contains $n-1$, then $a_{n-1} = n$. The number of such permutations is $(n-1)!$. If the excedance set contains $n-2$, then $a_{n-2} = n-1$ or n, so there are $2(n-1)!$ such permutations. Finally, if it contains $n-1$ and $n-2$, then $a_{n-2} = n-1$ and $a_{n-1} = n$, so there are $(n-2)!$ permutations. By PIE, the answer is $3(n-1)! - (n-2)!$.

Problem 7

Let $I(n)$ be the number of involutions on $[n]$. Every involution on $[n]$ either fixes n or pairs it with some $i \in [n-1]$. If it fixes n, the restriction to $[n-1]$ is still an involution, so there are $I(n-1)$ such involutions. If n is paired with some i, there are $n-1$ choices for i and, after choosing $\{i, n\}$, the remaining $n-2$ elements can be arranged in any involution, giving us $(n-1)I(n-2)$ possibilities. So

$$I(n) = I(n-1) + (n-1)I(n-2),$$

with $I(0) = 1$ and $I(1) = 1$. We show that $I(n)$ is even by induction. We have that $I(2) = 2$ and $I(3) = 4$, which are even. Now, assume $I(m)$ is even for all $2 \leq m \leq n-1$. For $n \geq 4$, $n-1$ and $n-2$ are ≥ 2, so $I(n-1)$ and $I(n-2)$ are even. The recurrence gives $I(n) = I(n-1) + (n-1)I(n-2)$, which is a sum of even numbers and is therefore even. So $I(n)$ is even, and the number of non-trivial involutions is $I(n) - 1$ which is odd.

Problem 8

Let X be finite and $f : X \to X$ a bijection. Then f decomposes into disjoint cycles, so each $x \in X$ lies in a unique cycle $x \to f(x) \to \cdots \to f^{\ell_x}(x) = x$ for some positive integer ℓ_x. Since X is finite, there are finitely many cycles and cycle lengths, say $\ell_1, \ldots, \ell_k$. Let $N = \operatorname{lcm}(\ell_1, \ldots, \ell_k)$. For any x, we have that n is a multiple of ℓ_x, so $f^n(x) = x$.

Problem 11

Let π be a derangement of $[n]$ and let $k = \pi(n)$. If $\pi(k) = n$, then π corresponds to a derangement of $[n] \backslash \{n, k\}$, since $\{n, k\}$ is a cycle and the remaining $n-2$ elements can be mapped freely. If $\pi(k) \neq n$, then create a derangement $\tau : [n-1] \to [n-1]$ by removing n and replacing it with k. Specifically, $\tau(i) = \pi(i)$ if $\pi(i) \neq n$, and $\tau(i) = k$ if $\pi(i) = n$. The total number of ways to do this is $(n-1)(d(n-1) + d(n-2))$.

We have that $d(1) = 0$, and using the formula, $d(1) = 1!(1-1) = 0$. Now, assume that $d(n) = n! \sum_{i=0}^{n} \frac{(-1)^n}{i!}$. Then

$$d(n+1) = n(d(n) + d(n-1))$$

$$= n\left(n! \sum_{i=0}^{n} \frac{(-1)^i}{i!} + (n-1)! \sum_{i=0}^{n-1} \frac{(-1)^i}{i!}\right) = n\left(n! \sum_{i=0}^{n} \frac{(-1)^i}{i!}\right) + n! \sum_{i=0}^{n-1} \frac{(-1)^i}{i!}$$

$$= (n+1)\left(n! \sum_{i=0}^{n} \frac{(-1)^i}{i!}\right) + (-1)^{n-1} = (n+1)! \sum_{i=0}^{n+1} \frac{(-1)^i}{i!},$$

as desired.